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SUMMARY

- Engineer building physics-based thermal, fluid and vehicle models in Python and ANSYS Fluent across scales, from multi-material heat-conduction solvers and 0.5 mm reacting-flow meshes to a 2 kg aircraft designed, built and flown.
- Trusts a model only after something independent has checked it, reflected in solvers validated against analytical solutions, CFD held to mesh independence and published experiment, and simulation models compared against instrumented flight hardware.

PROJECTS

ThermoSand: 2-D Heat-Conduction Solver and Interactive Thermal Sandbox *Python, NumPy, Streamlit | Summer 2026*

- **Built a 2-D heat-conduction solver in vectorized NumPy that converges 33x faster than its Jacobi baseline** (153 against 5064 iterations to the same tolerance), comprising: (1) a conservative finite-volume stencil with harmonic-mean face conductivities across 8 engineering materials, from copper and aluminum to alumina and cork, (2) fixed-temperature, fixed-flux and convective boundaries with volumetric heat sources, and (3) red-black successive over-relaxation at the theoretical optimum relaxation factor alongside an explicit transient integrator held to the mixed-material stability limit.
- **Validated the solver on a 12-test suite, matching a copper-ceramic series slab to round-off where the previous material-blind stencil was off by 96 K**, together with fixed-flux, convective-wall (Biot 1.67) and internal-heating cases against closed-form solutions, a 0.038 K RMS match to the 2-D series solution under an imposed 100 K difference, and energy conserved to round-off across a 257:1 conductivity interface.
- **Wrapped the solver in an interactive Streamlit app with 3 synchronized diagnostics** (Plotly temperature field, centerline profile, residual history), per-edge boundary control and steady and transient modes, rendering transient playback on one color scale held fixed across every frame so thermal evolution is read directly rather than through per-frame rescaling.

DroneCell Sim: Battery Pack Power and Resistive-Loss Simulator *JavaScript, React | In Progress*

- **Building a UAV battery endurance simulator that solves the constant-power load line at every timestep**, comprising: (1) an empirical open-circuit-voltage curve (Chen and Rincón-Mora), (2) cell internal resistance scaled from cell to pack, with series adding voltage and parallel dividing resistance, and (3) coulomb-counted state of charge to a 3.0 V cutoff.
- **Exposed a 64% endurance shortfall at 1 kW that energy-bucket models miss entirely** (2.9 against 8.0 min), with a fifth of the pack energy lost to resistive heating, against under 5% at 300 W (25.4 against 26.6 min), on a 6S2P pack of 3.0 Ah, 30 mΩ cells.
- **Surfaced the penalty through live series/parallel, cell-capacity, resistance and load controls**, reporting usable-versus-nameplate energy, resistive (I^2R) heat and peak C-rate for every configuration, enabling an engineer to watch the endurance penalty appear instead of being told about it.

Autonomous UAV: Mechanical, Electrical and Software Integration *Principal Architect and Flight Director, team of 4*

- **Designed, built and flew a 2 kg MTOW autonomous fixed-wing UAV as principal architect of a team of 4**, sized from first principles: NACA 2412 wing at 1.04 m span and aspect ratio 4.52 (8.4 kg/m² wing loading), conventional tail sized by volume coefficient, CG at a 25% static margin, and a bench-tested O.S. MAX 15LA glow engine.
- **Closed the loop from prediction to measurement across 5 instrumented flight tests**, building XFLR5, Python and Simulink models before first flight and comparing them against logged flight data, then rebuilt the empennage in reinforced aeroply after losing it to an aerobatic maneuver.
- **Integrated a 4-sensor avionics stack on a SpeedyBee STM32 flight controller running ArduPlane**: (1) NEO-6M GPS, (2) MPU-9250 IMU, (3) MPXV7002DP differential-pressure airspeed and (4) HC-12 telemetry, with parameters tuned for flight-plan enforcement, enabling autonomous waypoint navigation and loiter.

CFD Analysis of Airfoil Icing: CATIA-to-Fluent Pipeline *AE 269 Advanced CFD, SJSU | Team of 4*

- **Built a CATIA-to-Fluent CFD pipeline for a NACA 2412 section across rime, clear and horn ice at light and severe accretion**, comprising: (1) a 200-point, 1000 mm chord geometry carried by STEP into ANSYS Workbench, (2) ice shapes parameterized by leading-edge cap (8 to 20 mm, ± 15 to ± 20 deg span) and horn height (25 to 45 mm) against a clean baseline, and (3) wall-resolved meshes at y^+ near 1, grid-converged by a mesh-independence study before any result was reported.
- **Quantified up to 12% lift loss and over 115% drag increase against the clean baseline** across the accretion sweep, validated against published experimental icing data, enabling the ice shapes to be ranked by aerodynamic penalty.
- **Traced the drag rise and early stall to destruction of the leading-edge suction peak**, where horn ice forces early laminar-to-turbulent transition and large-scale separation into bluff-body flow.

- **Ranked 8 fuel-oxidizer combinations for detonation performance through NASA CEA**, sweeping 1 to 10 bar and 200 to 500 K to resolve Chapman-Jouguet states up to 306 atm and 4203 K, where a normalized pressure-to-temperature index identified propane-oxygen (0.1496) as the strongest candidate at roughly double hydrogen-oxygen (0.0728).
- **Built a transient reacting-flow CFD model of an unwrapped 300 by 75 mm annular combustor in ANSYS Fluent**, comprising: (1) a 0.5 mm structured quadrilateral mesh (skewness below 0.25, orthogonal quality above 0.9), (2) density-based Roe flux-difference splitting with a stiff finite-rate chemistry solver, and (3) CFL 0.7 time stepping at 0.125 microseconds.
- **Sustained a coherent 7.7 bar hydrogen-air detonation front matching published two-dimensional results**, driven by a custom pressure-sensitive injector UDF switching between blocked, subsonic and choked inflow, and diagnosed the decay of propane-oxygen as a resolution and boundary-condition limit rather than a chemistry one.

RELEVANT EXPERIENCE

Conceptual Design Engineering Intern, TATA Advanced Systems Ltd., Bengaluru, India

Apr 2024 - Aug 2024

- **Led aerodynamic trade studies on co-flow and counter-flow fluidic thrust vectoring** and authored internal research on circulation control for flapless flight, both Coanda-effect approaches to generating control forces without moving surfaces, presenting the analyses and their trade-offs to senior engineers in formal design reviews.
- **Extended MDAO frameworks (OpenMDAO, OpenAeroStruct, OpenConcept) to couple aerodynamics, propulsion and power in one vehicle-level model** for conceptual design of hybrid-electric and turboelectric aircraft, sizing battery packs against mission energy budgets and running range and endurance sweeps across configurations.
- **Built Python flight-mechanics and stability tooling for constrained-deployment UAV systems**, evaluating post-deployment dynamics and computing longitudinal and lateral-directional stability derivatives across configuration variations.

AI and Augmented Reality Engineer, Orange and Teal Pvt. Ltd., Bengaluru, India

Apr 2025 - Jul 2025

- **Architected and trained custom convolutional neural networks in PyTorch** for an internal system automating creative workflows, with modular retraining pipelines that kept model performance improving as the datasets evolved.
- **Deployed the pipeline from model output through to delivery infrastructure**, tuning accuracy, scalability and inference latency through controlled experiments.
- **Built an end-to-end pipeline integrating CRM data ingestion, customer behavior analytics, segmentation and automated prompt generation** to produce individually targeted output with dynamic offer logic.

EDUCATION

San Jose State University, San Jose, CA

Aug 2025 - May 2027 (Expected)

Master of Science in Aerospace Engineering | GPA: 3.83 | Coursework: Computational Fluid Dynamics, Advanced Aerodynamics, Flight Dynamics and Control, Numerical Methods

M.S. Project: CryoSand, an open trade-space sandbox for in-space cryogenic propellant storage, advised by Dr. Periklis Papadopoulos. Two-phase Fluent datasets generated on the SJSU/NASA HPC cluster; targeting AIAA SciTech or ICES and JOSS.

BMS College of Engineering, Bengaluru, India

May 2024

Bachelor of Engineering in Aerospace Engineering

SKILLS

Heat Transfer & Thermal Modeling: Steady and Transient Conduction | Conservative Finite-Volume Solvers (harmonic-mean material interfaces) | Convective, Fixed-Flux and Fixed-Temperature Boundaries | Volumetric Heat Sources | Mixed-Material Explicit Stability Limits | Analytical Verification and Validation | Resistive (I^2R) Heat Generation in Battery Packs | Cryogenic Thermal Analysis

CFD & CAD: ANSYS Fluent (steady and transient, density-based) | CATIA V5 | STEP-Based CAD-to-CFD Pipeline (Workbench, DesignModeler) | Structured and Wall-Resolved Meshing | Mesh-Independence Studies | UDF Development | HPC Cluster Runs

Test & Model Correlation: Prototype Build and Test | Wind Tunnel Surface-Pressure Instrumentation | Sensor Integration (IMU, GPS, differential pressure) | Telemetry and Log Analysis | Model-to-Test Comparison

Controls, Optimization & Hardware: Design Optimization (OpenMDAO) | Flight Dynamics and Control (graduate coursework) | Simulink | STM32 Flight Controller Integration | ArduPilot Parameter Tuning | Stability Derivatives

Programming & AI: Python (Advanced; NumPy, vectorized solvers) | MATLAB | C++ (Familiar) | PyTorch | JavaScript and React | Streamlit | Plotly | Git | Test Automation (pytest) | Linux | AI Coding Assistants in Solver Development, with output checked against analytical benchmarks